Gaussian Covariance Concavity and Simplex Optimality near Independence

Local review manuscript — 31 October 2026

Abstract

For every probability law ν with finite second moment, we prove that its maximal covariance with a centered Gaussian is a concave function of the Gaussian covariance matrix. For a non-degenerate simplex target the Hessian is negative definite at every positive definite covariance. The proof uses Gaussian integration on the strata of a translated simplex normal fan and an exact factorization through the simplex boundary maps. A weighted choice of cycle coordinates cancels the extra boundary term. Singular covariance lifting and approximation extend the result to arbitrary targets.

For every fixed number of cells, we deduce the balanced Gaussian first-moment inequality, its equality cases, and global centroid coercivity. Complete transfer arguments then give regular-simplex optimality on dimension-independent intervals adjacent to zero: a balanced maximum for positive correlation, and a minimum over pairs with arbitrary matching masses for negative correlation. The latter includes the single-partition problem without prescribed masses. The intervals depend on the number of cells. Full-correlation ranges remain open; no sharp multiway-cut hardness result is asserted.

Review status. This is a local manuscript for mathematical review. Its source lineage and internal checks are recorded in Appendix D; no external review, publication, or priority clearance is claimed.

Contents

1 Statements and conventions **2**

2 Covariance concavity **3**
2.1 The simplex dual and its strata . . . 3
2.2 Intrinsic Gaussian integration . . . 4
2.3 Facet matrices . . . 5
2.4 The chain factorization . . . 5
2.5 An admissible gauge and the cycle estimate . . . 6
2.6 The exact covariance Hessian . . . 6
2.7 Singular covariance, finite lifting, and limits . . . 7

3 The balanced endpoint and a global coercivity estimate **8**

4 Ambient dimensions and discrete consequences **9**
4.1 Existence and the positive problem . . . 9
4.2 Free matching masses and the negative problem . . . 9
4.3 The precise low-influence corollary . . . 10

A Common estimates for the two transfers **11**
A.1 The same nearest frame controls the mismatch . . . 11
A.2 A saturated positive gap and a damped negative gap . . . 11
A.3 Weighted moments and uniform tails . . . 12
A.4 The quadratic Mehler error . . . 13
A.5 Exact mass repair on fixed facet slabs . . . 13
A.6 Comparison on a fixed ball . . . 14

B The positive-correlation transfer **14**

C The negative transfer with free common masses **15**
C.1 Uniform localization in the nearest frame . . . 15
C.2 The mixed comparison retains the mass penalty . . . 15
C.3 Repair and absorption of the unequal masses . . . 16

D Proof lineage and review boundaries **16**

1 Statements and conventions

Write γ_N for standard Gaussian measure on $\mathbb{R}^N$ and $\mathcal{P}_2(\mathbb{R}^N)$ for probability laws with finite second moment. For $\nu \in \mathcal{P}_2(\mathbb{R}^N)$ set

$$\mathcal{F}_\nu(C) = \sup\{\mathbb{E}[X \cdot Y] : X \sim N(0, C),\ Y \sim \nu\}, \qquad C \geq 0.$$

The supremum is over all couplings. Covariance matrices are varied in their ordinary linear coordinates.

Theorem 1.1 (Covariance concavity)**.** *For every finite N and $\nu \in \mathcal{P}_2(\mathbb{R}^N)$, the function $C \mapsto \mathcal{F}_\nu(C)$ is concave on the positive semidefinite cone. Equivalently $C \mapsto W_2^2(N(0, C), \nu)$ is convex. If ν has $d+1$ affinely independent sites in $\mathbb{R}^d$, each of positive mass, then $\mathcal{F}_\nu$ is smooth on the positive definite cone and its Hessian is negative definite there.*

Fix $m \geq 3$, put $d = m - 1$, and identify $H = \{x \in \mathbb{R}^m : \sum_i x_i = 0\}$ isometrically with $\mathbb{R}^d$. Let

$$v_i = e_i - \frac{1}{m}\mathbf{1}, \quad \mu_m = \mathbb{E} \max_{1 \leq i \leq m} Z_i, \quad \alpha = \frac{\mu_m}{m-1}, \quad c_i = \alpha v_i, \quad w = \frac{\mu_m^2}{m-1},$$

where the Z_i are independent standard normals. The regular partition is $g_i(x) = \mathbf{1}_{\{v_i \cdot x = \max_j v_j \cdot x\}}$, with arbitrary choices on null tie hyperplanes. Symmetry gives $\mathbb{E}[X g_i(X)] = c_i$: the centroids are multiples of v_i and $\sum_i v_i \cdot \mathbb{E}[X g_i(X)] = \mu_m$ determines the multiplier. In larger dimensions the partition ignores the unused coordinates. With the noise notation below, every benchmark $S_\rho(g) = B_\rho(g, g)$ is evaluated in dimension $d = m - 1$ and is unchanged by adding unused Gaussian coordinates. When $N < d$, the comparison is with this d-dimensional benchmark, not with an assignment g in $\mathbb{R}^N$.

A *relaxed assignment* is a measurable map into $\Delta_m = \{t \in [0,1]^m : \sum_i t_i = 1\}$. It is balanced if $\mathbb{E} f_i = 1/m$. Write $b_i(f) = \mathbb{E}[X f_i(X)]$ and $W(f) = \sum_i \|b_i(f)\|^2$. Let $\mathcal{O}_m$ denote the ordered regular centroid orbit under orthogonal maps. Label permutations are already implemented by orthogonal symmetries of the simplex.

Theorem 1.2 (Balanced endpoint and coercivity)**.** *For every balanced relaxed assignment in any finite dimension, $W(f) \le w$. If $N \ge d$, equality holds exactly for regular simplex partitions, up to orthogonal maps, labels, and null changes. There is a finite $K_m \ge 1$, depending only on m, such that for all $N \ge d$,*

$$\operatorname{dist}_{\mathrm{F}}(b(f), \mathcal{O}_m)^2 \le K_m(w - W(f)). \tag{1}$$

For $N < d$ the endpoint benchmark is strict.

For $|\rho| < 1$ let $T_\rho u(x) = \mathbb{E}u(\rho x + \sqrt{1-\rho^2}Z)$ and define

$$B_\rho(f,h) = \sum_i \langle f_i, T_\rho h_i\rangle_{L^2(\gamma_N)}, \qquad S_\rho(f) = B_\rho(f,f).$$

Theorem 1.3 (Near independence)**.** *For each fixed $m \ge 3$ there are numbers $\rho_m^+ > 0$ and $r_m^- > 0$, independent of ambient dimension, with the following properties in every finite ambient dimension N.*

(i) If $0 < \rho \le \rho_m^+$ and f is balanced, then $S_\rho(f) \le S_\rho(g)$.

(ii) If $0 < r \le r_m^-$ and $\mathbb{E}f = \mathbb{E}h = p$ for an arbitrary common probability vector p, then $B_{-r}(f,h) \ge B_{-r}(g,g)$.

For $N \ge d$ equality in (i) has the regular-simplex classification. Equality in (ii) forces $p = (1/m, \dots, 1/m)$ and $f = h$ equal to one common regular simplex partition. For $N < d$ both benchmarks are strict. All statements include relaxed assignments.

In particular the negative single-assignment minimum has no imposed mass constraint. This is not a theorem optimizing a prescribed unequal mass vector. The two intervals can differ and may shrink with m; no numerical bound uniform in m is claimed. The covariance proof and endpoint argument are given in the main text. Appendices A–C give the complete fixed-dimensional transfers. Section 4 states the precise external dimension reductions and completes the ambient argument.

Related problems. Heilman [1, Section 8, Conjecture 3] asks whether balanced four-cell first-moment maximizers in $\mathbb{R}^3$ are simplicial conical partitions, and identifies quantitative stability as a route to larger-cell near-zero noise results. Theorem 1.2 supplies this structure and stability, and extends them to every $m \ge 3$. The introduction following Theorem 1.3 of Heilman's 23 October 2026 preprint [2] still describes the four-cell and higher Standard Simplex cases as open. That discussion distinguishes the unrestricted Propeller problem, which does not prescribe cell masses, from the Standard Simplex problem. It also distinguishes Gaussian mean-width comparisons with prescribed individual variances. Our endpoint prescribes cell masses and permits unequal centroid lengths; it does not assume those variance constraints. These comparisons locate the related problems and do not constitute an exhaustive priority assessment. The full-correlation ranges remain open.

2 Covariance concavity

2.1 The simplex dual and its strata

First let $\nu = \sum_{i=0}^d p_i \delta_{y_i}$, where the sites are affinely independent and $p_i > 0$. For $C > 0$,

$$\mathcal{F}_\nu(C) = \min_h \left\{ \mathbb{E}\max_i (y_i \cdot X + h_i) - \sum_i p_i h_i \right\}, \qquad X \sim N(0, C). \tag{2}$$

Indeed the maximum bounds every coupling. The objective is at least $\max h - \sum p_i h_i \geq (\min p_i)(\max h - \min h)$, so a minimum exists modulo common offsets. Ties have Gaussian measure zero. Differentiation in h_i makes the winning masses equal to p_i, and that winning assignment attains the bound.

Every offset class has a unique representation $h_i = c - y_i \cdot a$. The cells are therefore the target simplex's normal fan translated by a. For each nonempty vertex set S, let Γ_S be the locus where all labels in S tie for the maximum. It is a simplicial cone in an affine space of dimension $d+1-|S|$, based at a, with boundary facets $\Gamma_{S\cup\{k\}}$, $k \notin S$. At $C = I$ define

$$\mu_S = \int_{\Gamma_S} \gamma_d(x)\, d\mathcal{H}^{d+1-|S|}(x), \qquad m_S = \frac{1}{\mu_S} \int_{\Gamma_S} x\gamma_d(x)\, d\mathcal{H}^{d+1-|S|}(x),$$

and let P_S be the orthogonal projection onto its tangent space. All these masses are positive and all polynomial moments are finite. For the apex, $\mu_{\{0,\dots,d\}} = \gamma_d(a)$ and its mean is a.

The offset objective is the Gaussian convolution of the fixed support function $\max_i y_i \cdot x$, evaluated at $-a$, plus $(\sum_i p_i y_i) \cdot a$. Linear growth of the support function gives smooth dependence on $a, C > 0$ by differentiation of the Gaussian density. The offset Hessian is the complete graph Laplacian with positive facet weights $\mu_{\{i,j\}}/|y_i - y_j|$. It is invertible modulo constants. The implicit function theorem thus gives smooth minimizing offsets and a well-defined optimized Hessian.

2.2 Intrinsic Gaussian integration

For $i \in S$ and $k \notin S$, the inward normal within the stratum is

$$n_{S,k} = \frac{P_S(y_i - y_k)}{|P_S(y_i - y_k)|}.$$

Intrinsic integration by parts yields, for positive-dimensional strata,

$$\mu_S P_S m_S = \sum_{k\notin S} \mu_{S\cup\{k\}} n_{S,k}. \tag{3}$$

For an edge stratum $e = \{i, j\}$ a second integration gives

$$\mu_e(P_e - \operatorname{Cov}_e X) = \sum_{k\notin e} \mu_{e\cup\{k\}} n_{e,k}(m_e - m_{e\cup\{k\}})^T. \tag{4}$$

To verify the signs, the intrinsic Gaussian derivative is $-P_S x\gamma_d$, whereas the outward boundary normal is $-n_{S,k}$. In the second identity, integrate the derivative of $((x - m_e) \cdot u)\gamma_d$ in a tangent direction v. The two negative boundary signs produce the displayed product. Both means lie in the same affine facet, so their difference is tangent. Truncation by intrinsic balls produces only polynomial Gaussian tail terms on the artificial sphere, which vanish as the radius tends to infinity. Lower-dimensional intersections have zero boundary measure.

We use the classical convex-support Brascamp–Lieb consequence

$$\operatorname{Cov}_e X \leq P_e. \tag{5}$$

For a direct applicable formulation, Bobkov–Ledoux [3, Proposition 2.1] allows a smooth strictly convex potential on an open convex support. In orthonormal intrinsic coordinates the relative interior of the facet is such a support, and its Gaussian potential has Hessian I. Apply that proposition to linear functions; the facet boundary has zero intrinsic measure. Neither strictness nor invertibility of the covariance deficit is needed.

2.3 Facet matrices

Orient each edge $e = (i, j)$ with $i < j$ and put

$$d_e = y_i - y_j, \qquad \ell_e = |d_e|, \qquad w_e = \mu_e/\ell_e.$$

For every target triangle g let $A_g > 0$ be its Euclidean area and set $D_g = 2A_g/\mu_g$. Let l_k be the global affine barycentric coordinate of vertex k and write $v_k = -\nabla l_k$. Then

$$\sum_k v_k = 0, \qquad v_k \cdot (y_i - y_j) = \delta_{kj} - \delta_{ki}. \tag{6}$$

For $k \notin e$, v_k is tangent to Γ_e. The vectors $P_e(y_i - y_k)$ form a basis of that tangent space, with dual rows v_k^T.

Index the columns of N_e, T_e by triangles $g = e \cup \{k\}$ and define

$$N_e[:, g] = \frac{\mu_g}{\ell_e} n_{e,k}, \qquad T_e[:, g] = m_e - m_g, \qquad E_e = w_e(P_e - \operatorname{Cov}_e X).$$

All matrices and inverses in this paragraph act on the facet tangent space. The triangle height formula gives

$$N_e[:, g] = \frac{P_e(y_i - y_k)}{D_g}, \qquad \operatorname{row}_g(N_e^{-1}) = D_g v_k^T. \tag{7}$$

Equations (4)–(5) imply

$$E_e = N_e T_e^T = T_e N_e^T \geq 0, \quad T_e = E_e N_e^{-T}, \quad B_e = N_e^{-1} E_e N_e^{-T} = N_e^{-1} T_e \geq 0. \tag{8}$$

There is no inverse of E_e here. Equation (3) also gives $w_e P_e m_e = N_e \mathbf{1}$, hence the exact entries

$$B_e[g, g'] = \frac{1}{w_e} - D_g v_k \cdot m_{g'}, \qquad g = e \cup \{k\}. \tag{9}$$

2.4 The chain factorization

Let ∂_2 and ∂_3 be the usual boundary maps from oriented triangles to edges and from tetrahedra to triangles, on the complete simplex. All vertices in a simplex are ordered increasingly; for example $\partial_2[i, j, k] = [j, k] - [i, k] + [i, j]$. Coning chains to one fixed vertex proves that the graph cycle space is $\operatorname{im} \partial_2$ and $\ker \partial_2 = \operatorname{im} \partial_3$. In dimension two ∂_3 is absent.

Put $W = \operatorname{diag}(w_e)$ and $A = \partial_2^T W^{-1} \partial_2$. For a triangle coefficient vector λ, let $\eta_{e,g} = (\partial_2)_{e,g} \lambda_g$ on triangles incident to e, and define B by $\lambda^T B \lambda = \sum_e \eta_e^T B_e \eta_e$. For a target tetrahedron h, let V_h be its intrinsic three-dimensional volume, and set $c_h = \mu_h/(6V_h)$. Also write $D = \operatorname{diag}(D_g)$.

Lemma 2.1 (Exact boundary factorization)**.**

$$A - B = -D \partial_3 \operatorname{diag}(c_h) \partial_3^T D. \tag{10}$$

Proof. Suppose distinct triangles g, g' share edge e, and let $k = g \setminus e$. Because $k \notin g'$, the vector v_k is tangent to $\Gamma_{g'}$. Test (3) there against v_k. By (6), all boundary normals vanish in this pairing except the one adjoining k. Its pairing is the reciprocal target height $A_{g'}/(3V_h)$, where $h = g \cup g'$. Thus

$$v_k \cdot m_{g'} = \frac{\mu_h A_{g'}}{3 V_h \mu_{g'}}.$$

The off-diagonal entry of $A - B$ is therefore $(\partial_2)_{e,g}(\partial_2)_{e,g'} D_g D_{g'} c_h$. The identity $\partial_2 \partial_3 = 0$, on the shared edge of this tetrahedron, gives

$$(\partial_2)_{e,g}(\partial_2)_{e,g'} = -(\partial_3)_{g,h}(\partial_3)_{g',h}.$$

This proves the claimed off-diagonal entries. Triangles sharing no edge interact in neither matrix.

For the diagonal, summing (9) over the three edges of g and using (6) gives

$$(A - B)_{g,g} = D_g \sum_{k \in g} v_k \cdot m_g = -D_g \sum_{l \notin g} v_l \cdot m_g = -D_g^2 \sum_{h \supset g} c_h.$$

The last equality uses the same stratum mean identity and reciprocal height. If $d = 2$, the sum over outside vertices is empty and the difference is zero, also when the triangle stratum is the apex. These are all entries of (10). The volumes in this argument are intrinsic to the target tetrahedra, so there is no factor depending on the ambient dimension. □

2.5 An admissible gauge and the cycle estimate

For a fixed edge cycle j, choose $\partial_2 \lambda = j$. Its representatives form $\lambda + \operatorname{im} \partial_3$. Minimize $\lambda^T D \lambda$ on this closed affine space. Since $D > 0$, the minimizing representative exists and satisfies

$$\partial_3^T D \lambda = 0. \tag{11}$$

This uses no injectivity of ∂_3. It changes only cycle coordinates. By (10),

$$\lambda^T B \lambda = \lambda^T A \lambda = j^T W^{-1} j. \tag{12}$$

Give the cells arbitrary vectors z_i and write $z_e = z_i - z_j$, $a_e = z_e \cdot m_e$. Signed edge differences telescope around each triangle, so

$$j \cdot a = \sum_e z_e \cdot T_e \eta_e.$$

By (8), square-root Cauchy–Schwarz in the PSD matrices E_e, followed by Cauchy–Schwarz across edges, gives

$$(j \cdot a)^2 \le \left(\sum_e z_e^T E_e z_e \right) j^T W^{-1} j.$$

Singular deficits cause no exception. Weighted least-squares duality on the edge graph now yields

$$\min_b \sum_e w_e (a_e + b_i - b_j)^2 \le \sum_e w_e \big(|P_e z_e|^2 - \operatorname{Var}_e(z_e \cdot X) \big). \tag{13}$$

For clarity, at the least-squares minimizer the weighted residual is a cycle. Cauchy–Schwarz gives the dual supremum $\sup_{j \neq 0} (j \cdot a)^2 / (j^T W^{-1} j)$, and that residual attains it unless it is zero. Thus (13) uses a single global offset vector b, not offsets chosen independently on facets.

2.6 The exact covariance Hessian

Take $C(t) = I + tH$ with $H = H^T$. The first two derivatives of its square root at zero are $H/2$ and $-H^2/4$. Differentiating the affine maximum in (2) and eliminating the offset derivative gives

$$4D^2 \mathcal{F}_\nu(I)[H, H] = \min_b \sum_e w_e \mathbb{E}[(X \cdot H d_e + b_i - b_j)^2 \mid \Gamma_e] - \sum_e w_e |H d_e|^2. \tag{14}$$

Here conditional expectation means the normalized intrinsic facet measure. To detail the second variation, the score velocity difference across a facet is $X \cdot Hd_e/2 + h'_i - h'_j$. Differentiating the first derivative over moving cells contributes its square divided by ℓ_e, integrated over the facet. Each unoriented facet is counted once. Gaussian integration by parts for the winning vector gives $\mathbb{E}[XY(X)^T] = \sum_e w_e d_e d_e^T$, so the interior acceleration is $-\frac{1}{4}\sum_e w_e |Hd_e|^2$. Put $b = 2h'$. The h'' term vanishes because the current masses equal p. The positive offset Hessian makes the displayed minimum exactly the Schur complement. Polyhedral truncation and Gaussian tails justify the boundary differentiation; multiple ties have zero facet measure, and the smoothness already proved identifies this calculation with the ordinary Hessian.

Apply (13) with $z_i = Hy_i$. The facet variances cancel, leaving

$$4D^2\mathcal{F}_\nu(I)[H,H] \le -\sum_e w_e \frac{(d_e^T H d_e)^2}{\ell_e^2}. \tag{15}$$

The edge dyads of an affinely independent simplex span the symmetric matrices: differences $b_i = y_i - y_0$ form a basis, and the quadratic forms at b_i and $b_i - b_j$ recover every bilinear entry in that basis. Thus the right side is strictly negative for $H \neq 0$. At any $C_0 > 0$, transform the target sites by $C_0^{1/2}$ and the covariance direction K by $H = C_0^{-1/2} K C_0^{-1/2}$. This reduces to the calculation at I, without a commutativity assumption. Integration along segments proves concavity on the positive definite cone. For $d = 1$ the value is a positive constant times the square root of variance, giving the same assertions.

2.7 Singular covariance, finite lifting, and limits

For every fixed $\nu \in \mathcal{P}_2$,

$$|\mathcal{F}_\nu(C) - \mathcal{F}_\nu(C')| \le (\mathbb{E}|Y|^2)^{1/2} \|C^{1/2} - (C')^{1/2}\|_{\mathrm{F}}. \tag{16}$$

Indeed any coupling of $C^{1/2}G$ with Y lifts to one with standard G, using its Gaussian conditional law even when C is singular. Reuse G, Y for C' and apply Cauchy–Schwarz; then reverse the argument. Adding εI and taking limits extends simplex concavity to the PSD cone.

For an arbitrary finite target $\sum_{i=1}^m p_i \delta_{y_i}$ in $\mathbb{R}^N$, lift its sites to $(y_i, \varepsilon e_i)$ in $\mathbb{R}^{N+m}$, for any fixed $\varepsilon > 0$. They are affinely independent. A Gaussian with covariance $\operatorname{diag}(C, 0)$ has exactly the old cost, so the two optimal values coincide. Subtract a common lifted site; its expected contribution is zero. Project onto the $(m-1)$-dimensional span S of lifted differences. The projected source has covariance $P_S \operatorname{diag}(C, 0) P_S$, linear in C, and its cost is unchanged. Every projected coupling lifts by the source conditional law. Simplex PSD concavity therefore pulls back to the original finite-target functional. A one-atom target has value zero.

Finally, gluing a coupling of two targets to a Gaussian-target coupling gives

$$|\mathcal{F}_\nu(C) - \mathcal{F}_{\nu'}(C)| \le \sqrt{\operatorname{tr} C}\, W_2(\nu, \nu'). \tag{17}$$

Finite laws are dense in $\mathcal{P}_2$, by truncation and quantization. Pass to the limit in the concavity inequality. The identity $W_2^2(N(0,C),\nu) = \operatorname{tr} C + \int |y|^2 \, d\nu - 2\mathcal{F}_\nu(C)$ proves the equivalent assertion in Theorem 1.1. Neither lifting nor limits assert strictness for general targets.

Corollary 2.2. *The function $C \mapsto \mathcal{F}_\nu(C)^2$ is concave on the PSD cone.*

Proof. Independence gives $\mathcal{F}_\nu \ge 0$, and scaling the Gaussian gives $\mathcal{F}_\nu(tC) = \sqrt{t}\,\mathcal{F}_\nu(C)$. For $0 < a < 1$, concavity applied to C/a and $D/(1-a)$ yields

$$\mathcal{F}_\nu(C + D) \ge \sqrt{a}\,\mathcal{F}_\nu(C) + \sqrt{1-a}\,\mathcal{F}_\nu(D).$$

Optimizing a, with endpoints obtained by limits, proves superadditivity of $\mathcal{F}_\nu^2$. Its degree-one homogeneity then proves concavity. □

3 The balanced endpoint and a global coercivity estimate

Let $P = I - \mathbf{1}\mathbf{1}^T/m$, $K_* = mP$, and $L_0 = m(m-1)$. On zero-sum score covariances define

$$J_m(K) = \inf_{\sum_i h_i = 0} \mathbb{E} \max_i (X_i + h_i), \qquad X \sim N(0, K), \quad K \geq 0,\ K\mathbf{1} = 0.$$

This is the maximal covariance functional for the uniform regular target $v_1, \dots, v_m$ on the zero-sum space. The dual remains valid for singular K by continuity. Indeed the difference between two expected maxima, coupled through one standard Gaussian and with the same offsets, is at most $\|K^{1/2} - (K')^{1/2}\|_{\mathrm{F}}$, uniformly in those offsets. The same bound therefore controls their infima. Approximate on the zero-sum space by positive definite covariances and also use (16). The assignment inequality below follows directly from the pointwise maximum, irrespective of singular dual attainment.

On $\operatorname{tr} K = L_0$, permutation averaging gives K_*: invariance forces constant diagonal and off-diagonal entries, fixed by the trace and zero row sums. Theorem 1.1 and permutation invariance therefore imply

$$J_m(K) \leq J_m(K_*) = \sqrt{m}\,\mu_m = \sqrt{L_0 w}. \tag{18}$$

At K_* the winner assignment is balanced by symmetry, so zero offsets attain this value.

The strict Hessian in (15), continuity, and permutation symmetry give constants $c_0, \delta_0 > 0$ such that

$$J_m(K_*) - J_m(K) \geq c_0 \|K - K_*\|_{\mathrm{F}}^2 \quad \text{when } \|K - K_*\|_{\mathrm{F}} \leq \delta_0, \quad \operatorname{tr} K = L_0.$$

Indeed the gradient at K_* is scalar on the zero-sum space, so the first derivative vanishes on trace-zero directions; integrate the negative Hessian in a small neighborhood. The trace-normalized PSD domain is compact. Choose one $\theta \in (0,1)$ so that $K_\theta = (1-\theta)K_* + \theta K$ lies in that neighborhood for every K in the domain. Concavity gives

$$\theta[J_m(K_*) - J_m(K)] \geq J_m(K_*) - J_m(K_\theta) \geq c_0\theta^2 \|K - K_*\|_{\mathrm{F}}^2.$$

Thus, with $\kappa_m = c_0\theta > 0$,

$$J_m(K_*) - J_m(K) \geq \kappa_m \|K - K_*\|_{\mathrm{F}}^2 \tag{19}$$

on the entire normalized PSD domain, including its boundary.

For balanced relaxed f with $W = \sum_i |b_i|^2 > 0$, normalize $B_i = \sqrt{L_0/W}\, b_i$ and let $K = BB^T$. Since $\sum_i b_i = 0$, $K\mathbf{1} = 0$ and $\operatorname{tr} K = L_0$. The same assignment is an admissible balanced score coupling, hence

$$\sqrt{L_0 W} = \mathbb{E} \sum_i f_i(X) B_i \cdot X \leq J_m(K) \leq \sqrt{L_0 w}.$$

This proves $W \leq w$; $W = 0$ is immediate. If $W = w$, (19) forces $K = K_*$. Equality in the pointwise maximum inequality then forces f to select the unique regular winning score almost everywhere. This also proves the relaxed equality classification. In $N < d$, pad the source by independent Gaussian coordinates; the centroid Gram has rank at most N and cannot equal a regular rank-d Gram, proving strictness.

For quantitative centroid alignment, put $\Delta = w - W$. The difference of square roots and (19) imply

$$\|K - K_*\|_{\mathrm{F}}^2 \leq \frac{\sqrt{L_0}}{\kappa_m \sqrt{w}} \Delta.$$

For $N \geq d$, orthogonal Procrustes alignment with any regular row matrix B_* of Gram K_* gives

$$\begin{aligned}\min_{O \in O(N)} \|B - B_* O\|_{\mathrm{F}}^2 &= \sum_{j=1}^{m-1} (\sqrt{\lambda_j(K)} - \sqrt{m})^2 \\ &\leq \frac{1}{m} \|K - K_*\|_{\mathrm{F}}^2.\end{aligned}$$

Rescale the two matrices by $\sqrt{W/L_0}$ and $\sqrt{w/L_0}$ respectively. The two-square inequality, $W \leq w$, and $(\sqrt{W} - \sqrt{w})^2 \leq \Delta$ show that

$$\mathrm{dist}_{\mathrm{F}}(b, \mathcal{O}_m)^2 \leq \left(2 + \frac{2\sqrt{w}}{m\kappa_m \sqrt{L_0}}\right) \Delta. \tag{20}$$

The case $W = 0$ is immediate with this coefficient. This proves Theorem 1.2, with K_m equal to the displayed coefficient or any larger number. It is finite and independent of N; no explicit numerical value or uniform bound as $m \to \infty$ is asserted.

4 Ambient dimensions and discrete consequences

The proofs in Appendices B and C take place in $d = m - 1$. We use the following two external results with different feasible classes.

Heilman–Tarter [4, Theorem 1.9] states that, for positive correlation and prescribed positive cell masses, every maximizing Gaussian partition in dimension $N \geq m - 1$ depends, after a common orthogonal change of coordinates, on at most $m - 1$ coordinates. Heilman [5, Problem 7.1 and Theorem 7.9] instead treats minimizers of a positive-kernel bilinear objective over two partitions whose corresponding masses agree but are *free*. Every such minimizer has a common $(m - 1)$-dimensional reduction. This latter result has no hyperstability hypothesis and is not the distinct theorem for two separately prescribed mass vectors.

4.1 Existence and the positive problem

For finite N, the simplex-valued feasible set with prescribed masses is weakly compact in $L^2(\gamma_N)^m$. The Ornstein–Uhlenbeck operator is compact: each Hermite degree has finite multiplicity and $|\rho|^k \to 0$. Thus the positive quadratic objective is weakly continuous on this bounded set and attains its maximum.

Every maximizer is an indicator partition. Otherwise two fixed labels are both uniformly fractional on a positive-measure set. Choose two disjoint subsets of equal positive measure there. Opposite label perturbations on those subsets preserve all masses and produce two distinct feasible assignments whose average is the original one. Strict positivity of T_ρ for $\rho > 0$, visible from its strictly positive Hermite eigenvalues, makes the average of their objectives strictly larger. This contradicts maximality. Theorem 1.9 of [4] therefore applies to every relaxed maximizer. Appendix B gives its value and its exact equality form.

4.2 Free matching masses and the negative problem

The set of pairs with free common masses is weakly compact, and compactness of T_{-r} makes the bilinear objective jointly weakly continuous on it. A minimizing pair exists. Hold its attained mass vector p fixed. With h fixed, the minimizing face for the continuous linear functional in f has an extreme point. Every extreme point of this fixed-mass simplex-assignment set is an

indicator, by the same equal-measure opposite perturbation construction. Choose such an f_1, and then with f_1 fixed choose an indicator minimizer h_1. The pair has the same global minimum. This sequential argument does not require an incorrect general principle about extreme points of bilinear optimization.

The competitive bounds (33) hold in every finite ambient dimension, and (37) makes all masses positive. Reflect the second partition: $B_{-r}(f_1, h_1) = B_r(f_1, h_1(-\cdot))$. The free-common-mass theorem [5, Theorem 7.9] now reduces this indicator minimizer to a common subspace of dimension at most d. Reflection back preserves that subspace. Appendix C forces $f_1 = h_1 = g$ in one common frame.

For the original fractional pair, the intermediate pair (g, h) still has the minimum value. Its excess over (g, g) is exactly $D_-(h)$. Strict reference ordering forces $h = g$. The original pair (f, g) then has excess $D_-(f)$, so $f = g$. This recovers equality for the original fractional pair, not only for an indicator representative. In either sign, an assignment in $N < d$ can be embedded in dimension d; equality would require a regular centroid Gram of rank d, impossible for the embedded rank at most N. This completes Theorem 1.3.

4.3 The precise low-influence corollary

On $[m]^n$ use uniform product measure and the noise channel which, in each coordinate, keeps a symbol with probability $(1 + (m-1)\rho)/m$ and changes it to each particular different symbol with probability $(1-\rho)/m$. It exists for $-1/(m-1) \le \rho \le 1$. For $f : [m]^n \to \Delta_m$ define

$$\mathrm{Inf}_k(f) = \sum_i \mathbb{E}\,\mathrm{Var}(f_i \mid \text{all coordinates except } k).$$

Write $\mathrm{Stab}_\rho(f) = \sum_i \mathbb{E}[f_i(X) f_i(Y)]$ for this channel, and $p_m(\rho) = S_\rho(g)$.

Corollary 4.1. *Fix $m \ge 3$, an admissible nonzero ρ in the corresponding interval of Theorem 1.3, and $\varepsilon > 0$. There is $\tau > 0$, depending on m, ρ, ε, such that for every n and every f with $\max_k \mathrm{Inf}_k(f) \le \tau$,*

$$\mathrm{Stab}_\rho(f) \le p_m(\rho) + \varepsilon \quad (\rho > 0,\ \mathbb{E}f = \mathbf{1}/m),$$

whereas

$$\mathrm{Stab}_\rho(f) \ge p_m(\rho) - \varepsilon \quad (\rho < 0,\ \textit{with arbitrary } \mathbb{E}f).$$

Proof. Isaksson–Mossel [6, Theorem 7.1] transfers a low-influence simplex-valued function to a Gaussian relaxed assignment with the same expectation vector and approximates stability at the same fixed ρ. Apply Theorem 1.3, using its free-mass single-assignment consequence for the negative case, and allocate the approximation error below ε. Lemma 7.3 of [6] identifies $p_m(\rho)$ with the limiting plurality stability. This gives the stated additive comparison. □

There is no uniform-in-ρ influence threshold or assertion of exact finite-electorate optimality. Only the single-rule arbitrary-bias negative corollary is asserted here; no additional discrete bilinear theorem is needed. The near-zero Gaussian intervals do not give a full-range Standard Simplex theorem or a sharp MAX-m-CUT hardness ratio.

A Common estimates for the two transfers

Throughout the transfer appendices $d = m-1$, the reference g and centroids c_i are as in Section 1, and $K = K_m \geq 1$ satisfies (1). Set

$$a = |c_i - c_j| = \sqrt{2}\alpha, \qquad P_0 = \binom{m}{2}, \qquad \beta = P_0\sqrt{2/\pi}/a, \qquad \sigma = (K+1)^{-1}.$$

The first moment obeys $w \leq 1$ by Parseval, so $a^2 = 2w/(m-1) \leq 1$. All integrals in these appendices use γ_d unless specified otherwise.

A.1 The same nearest frame controls the mismatch

For balanced f choose a nearest regular centroid matrix c and its partition g. Put $\Delta = w - W(f)$, $q = \|b(f) - c\|_{\mathrm{F}}^2$, $\eta = \frac{1}{2}\sum_i \|f_i - g_i\|_1$, and $J = w - \sum_i c_i \cdot b_i(f)$. Then

$$2J = \Delta + q \leq (1+K)\Delta. \tag{21}$$

If Γ is the gap between the two largest regular linear scores, a union bound on the P_0 Gaussian score differences of standard deviation a gives $\mathbb{P}(\Gamma \leq t) \leq \beta t$. With $p_{\mathrm{wrong}} = 1 - f_{\mathrm{winner}}$, $J \geq \mathbb{E}[p_{\mathrm{wrong}}\Gamma]$ and $\mathbb{E}p_{\mathrm{wrong}} = \eta$. Layer cake yields

$$J \geq \int_0^\infty (\eta - \beta t)_+ \, dt = \frac{\eta^2}{2\beta}, \qquad \eta^2 \leq L\Delta, \quad L = \beta(1+K).$$

Thus no second choice of alignment is needed.

A.2 A saturated positive gap and a damped negative gap

Let ϕ, Φ be the standard normal density and distribution function, and

$$H_\delta(u) = \phi(u)\Phi(u+\delta) - \phi(u+\delta)\Phi(u).$$

For $\delta > 0$, this is positive: $\Phi(u)/\phi(u)$ is strictly increasing, as is clear from its integral representation $\int_0^\infty e^{uy - y^2/2}\, dy$. In reference cell i, put $z_{ij} = (v_i - v_j)\cdot x/\sqrt{2} \geq 0$. For independent normal utilities with label i of mean zero and label j of mean $-\delta$, the winning-probability difference is

$$\mathbb{E}\left[\int_M^\infty H_\delta(u)\, du\right],$$

where M is the maximum of the other utilities. Lowering their means increases this difference.

For $G_i = T_r g_i$, subtract the mean of winner i. The other means are nonpositive and $\delta = \sqrt{2}\, r z_{ij}/\sqrt{1-r^2}$. Raise all other means to zero; the event $M \leq 0$ has probability $2^{-(m-2)}$. For $0 \leq u, \delta \leq 1$, $\partial_\delta H_\delta(u) \geq \phi(1)\phi(2)$. Integrate over $0 \leq u \leq 1$; for larger δ use monotonicity when the competitor mean is lowered. Consequently

$$G_i - G_j \geq c_+ \min(r z_{ij}, 1), \qquad c_+ = \frac{e^{-5/2}}{2^{m-1}\pi}. \tag{22}$$

For $G_i = T_{-r} g_i$, subtract label j's mean instead. Now j has mean zero, i has mean $-\delta$, and every other mean is at most $U_x = \sqrt{2}\, r|x|/\sqrt{1-r^2}$. Raise those means to U_x. Their maximum is at most U_x with probability $2^{-(m-2)}$. On $U_x \leq u \leq U_x + 1$ and $0 \leq \delta \leq 1$,

$$\partial_\delta H_\delta(u) = \phi(u)\phi(u+\delta) + (u+\delta)\phi(u+\delta)\Phi(u) \geq \phi(U_x+1)\phi(U_x+2).$$

The same integral argument and monotonicity for $\delta \geq 1$ give a gap at least $c_+ e^{-U_x^2 - 3U_x} \min(\delta, 1)$. Since $3U_x \leq U_x^2 + 9/4$ and $2U_x^2 = 4r^2|x|^2/(1-r^2) \leq |x|^2/32$ for $r \leq 1/16$,

$$G_j - G_i \geq c_- \min(r z_{ij}, 1) e^{-|x|^2/32}, \qquad c_- = \frac{e^{-19/4}}{2^{m-1}\pi}. \tag{23}$$

Thus the reference winner is the unique largest positive potential and the unique smallest negative potential, outside null hyperplanes.

A.3 Weighted moments and uniform tails

Let

$$\psi(x) = e^{|x|^2/8}, \qquad \omega_r(x) = \min\left(r \min_{i<j} |(v_i - v_j) \cdot x|/\sqrt{2},\, 1\right).$$

Use $\widetilde{\omega}_r^+ = \omega_r$ and $\widetilde{\omega}_r^- = e^{-|x|^2/32}\omega_r$. Define the finite constants

$$B_+ = P_0 \left[2^{(d-1)/2}\phi(0)e^{1/4} + 2^{d/2-2}\right],$$
$$B_- = P_0 \left[\phi(0)e^{5/16}(8/3)^{(d-1)/2} + \tfrac{1}{4}(8/3)^{d/2}\right].$$

For $|h| \leq 1$, $\lambda > 0$, and $0 < r \leq 1$,

$$\int |h|\psi \leq \lambda \int |h|\widetilde{\omega}_r^{\pm} + \frac{B_\pm}{\lambda r}. \tag{24}$$

Here are the constants explicitly. For the positive case dominate the positive part by summing over the individual planes. Write $x = ue + z$. On $|u| \leq 1$, replace ψ by $A(z) = e^{1/8}e^{|z|^2/8}$ and $\phi(u)$ by $\phi(0)$, and integrate $(A - \lambda r|u|)_+$ over the whole line. Its integral is $A^2/(\lambda r)$. On $|u| > 1$ the saturated plane weight is at least r, and $(A-B)_+ \leq A^2/(4B)$. Gaussian exponential moments give B_+. For the negative case use the exact identity

$$(\psi - \lambda\widetilde{\omega}_r^-)_+ = e^{-|x|^2/32}(e^{5|x|^2/32} - \lambda\omega_r)_+ \leq (e^{5|x|^2/32} - \lambda\omega_r)_+.$$

The same calculation, with this auxiliary exponential, gives B_-. The moment controlled on the left of (24) remains ψ.

For any relaxed k, define the nonnegative losses

$$D_+(k) = -\sum_i \langle k_i - g_i, T_r g_i\rangle, \qquad D_-(k) = \sum_i \langle k_i - g_i, T_{-r} g_i\rangle.$$

Each vanishes only for $k = g$ by the strict gaps. Optimizing (24), using $\sum_i |k_i - g_i| = 2p_{\text{wrong}}$ and (22) or (23), gives

$$\sum_i \left(\int |k_i - g_i|\psi\right)^2 \leq M_\pm D_\pm(k)/r, \qquad M_\pm = 8B_\pm/c_\pm. \tag{25}$$

In the relevant sign put $A_\pm^2 = \max(1, m/4)M_\pm$. If $z = \sqrt{D_\pm(k)/r}$ and $\eta = \frac{1}{2}\sum_i \|k_i - g_i\|_1$, then $\eta \leq A_\pm z$. For $R \geq 2$, let $\varepsilon_R = (1+R)e^{-R^2/8}$. The pointwise bound $(1+|x|)\mathbf{1}_{\{|x|>R\}} \leq \varepsilon_R \psi(x)$ implies both

$$\frac{1}{2}\sum_i \left|\mathbb{E}[(k_i - g_i)\mathbf{1}_{\{|X|>R\}}]\right| \leq \varepsilon_R A_\pm z, \qquad \left(\sum_i \left|\mathbb{E}[X(k_i - g_i)\mathbf{1}_{\{|X|>R\}}]\right|^2\right)^{1/2} \leq \varepsilon_R A_\pm z. \tag{26}$$

These are separate bounds, obtained from Cauchy–Schwarz across labels.

A.4 The quadratic Mehler error

The Mehler kernel relative to $\gamma_d \otimes \gamma_d$ is

$$M_\rho(x,y) = (1-\rho^2)^{-d/2} \exp\left(\frac{2\rho x\cdot y - \rho^2(|x|^2+|y|^2)}{2(1-\rho^2)} \right).$$

For $|\rho| \le 1/16$, writing $\ell = \log M_\rho$ and $L_{xy} = |x|^2 + |y|^2$, direct differentiation gives $|\ell'| \le d + L_{xy}$, $|\ell''| \le 2d + 2L_{xy}$ and $M_\rho \le 2^d e^{L_{xy}/34}$. For example the nondimensional parts of ℓ' and ℓ'' are

$$\frac{(1+\rho^2)x\cdot y - \rho L_{xy}}{(1-\rho^2)^2}, \qquad \frac{2\rho(3+\rho^2)x\cdot y - (1+3\rho^2)L_{xy}}{(1-\rho^2)^3};$$

$|x\cdot y| \le L_{xy}/2$ gives the stated bounds. The exponent bound follows from $|\rho|/[2(1+|\rho|)] \le 1/34$. Put $t_0 = 13/136$ and

$$C_d = 2^{d-1}\left[d^2 + 2d + \frac{2d+2}{2t_0} + \frac{1}{t_0^2} \right].$$

Taylor's theorem, $M'' = M((\ell')^2 + \ell'')$, and the maxima of $L_{xy}e^{-t_0 L_{xy}}$ and $L_{xy}^2 e^{-t_0 L_{xy}}$, using $e > 2$, give

$$|M_\rho - 1 - \rho x\cdot y| \le C_d \rho^2 \psi(x)\psi(y). \tag{27}$$

Thus a mixed remainder for deviations $f-g, h-g$ is bounded by $C_d M_\pm r\sqrt{D_\pm(f)D_\pm(h)}$.

A.5 Exact mass repair on fixed facet slabs

Fix a reference frame, $R \ge 3$, and a relaxed k with mismatch η. Let $p = \mathbb{E}k$, $U = \mathbf{1}/m$, $u = \frac{1}{2}\|p - U\|_1$. Truncate k to g outside B_R, obtaining k^R, whose half-ℓ^1 mass imbalance v satisfies

$$v \le u + \varepsilon_R A_\pm z, \qquad v \le \eta.$$

Set

$$c_A = \mathrm{Vol}(B_{d-1})\, 8^{-(d-1)}(2\pi)^{-d/2}e^{-2} > 0.$$

For donor i and recipient j, use the facet disk centered at $v_i + v_j$, of tangential radius $1/8$, and its donor-side normal slab of width $\tau = (\eta + 2v)/c_A$. Assume $\eta \le c_A/24$, so $\tau \le 1/8$. At the anchor, the *unscaled* v_i and v_j scores tie and exceed all other scores by one. A displacement of size at most $\sqrt{2}/8$ reduces these gaps by at most $1/4$. Distinct unoriented anchors are at least $\sqrt{2}$ apart; opposite slabs at one anchor have disjoint interiors. All oriented supports therefore lie in their correct cells, are mutually disjoint up to null sets, and are contained in B_2.

Their Gaussian mass is at least $c_A\tau$. Since the total missing donor mass in its reference cell is at most η, each slab has available k_i^R mass at least $c_A\tau - \eta = 2v$. Choose any excess-to-deficit flow of total mass v, and transfer each pair mass proportionally to k_i^R on its slab. Each pair amount is at most v. Disjoint supports keep the assignment simplex-valued, and all masses are repaired exactly. If $v = 0$, make no transfers. Call the result k'.

Let $b' = b(k')$, $e_0 = \|b' - b(k)\|_{\mathrm{F}}$, and

$$J' = w - \sum_i c_i \cdot b_i', \qquad J_{\mathrm{in}} = \mathbb{E}\left[\mathbf{1}_{B_R} \sum_i k_i(X)(\max_j c_j\cdot X - c_i\cdot X) \right].$$

The linear cost per unit transferred mass is at most $a\tau \le \tau$. Its total is at most $3\eta v/c_A$. Tail removal and the centroid cost of transferring mass in B_3 consequently give

$$J' \le J_{\mathrm{in}} + \frac{3}{c_A}\eta u + C_J\varepsilon_R z^2, \qquad e_0 \le 6(u + \varepsilon_R A_\pm z), \qquad C_J = 3A_\pm^2/c_A. \tag{28}$$

For the first inequality use $v \le u + \varepsilon_R A_\pm z$ and $\eta \le A_\pm z$; for the second use (26) and $3\sqrt{2}v$ as the Frobenius centroid cost.

Suppose now that c is a nearest regular centroid matrix for $b(k)$, and put $q = \|b(k) - c\|_{\mathrm{F}}^2$. Distance to $\mathcal{O}_m$ is one-Lipschitz. Applying (1) to balanced k' gives $2J' \ge (1 + 1/K)(\sqrt{q} - e_0)_+^2$. Young's inequality with parameter $1/(2K)$, also covering $\sqrt{q} \le e_0$, therefore gives

$$q \le (2 - \sigma)J' + (2K + 1)e_0^2. \tag{29}$$

The linear cost retains the original frame; no new alignment is introduced.

A.6 Comparison on a fixed ball

Write $H_i(y) = \mathbb{E}g_i(y + Z)$. Then $\nabla H_i(0) = c_i$ and $\|\nabla^2 H_i\|_{\mathrm{op}} \le \sqrt{2}$, by Gaussian differentiation: in a unit direction the second derivative is $\mathbb{E}[g_i(y + Z)((u \cdot Z)^2 - 1)]$, bounded by Cauchy–Schwarz. Conicality gives $T_{\pm r}g_i(x) = H_i(\pm tx)$, $t = r/\sqrt{1 - r^2}$. On B_R and for $r \le 1/16$,

$$\|\nabla(T_{\pm r}g_i - 1/m \mp rc_i \cdot x)\| \le (1 + 3R)r^2.$$

The orthogonal reflection swapping vertices i, j fixes their bisector and all other vertices, so the pair remainder vanishes on the bisector. Integrating normally from that plane, inside B_R, gives the appropriate signed potential gap at least $r(c_i - c_j) \cdot x(1 - \beta_R r)$ in reference cell i, where

$$\beta_R = \frac{2(1 + 3R)}{a}, \qquad rJ_{\mathrm{in}} \le \frac{D_\pm(k)}{1 - \beta_R r} \quad \text{if } \beta_R r < 1. \tag{30}$$

B The positive-correlation transfer

Work first in dimension $d = m - 1$. Call balanced f competitive if $S_\rho(f) \ge S_\rho(g)$, and choose a nearest reference centroid frame. Write $D = D_+(f)$, $z = \sqrt{D/\rho}$, and use the positive constants from Appendix A. If $\Delta = w - W(f) = 0$, the endpoint equality already gives $f = g$, so assume otherwise.

Hermite expansion, positivity of its coefficients for $\rho > 0$, and Parseval give for a competitive assignment

$$\frac{1}{m} + \rho w \le S_\rho(g) \le S_\rho(f) \le \frac{1}{m} + \rho W(f) + \rho^2(1 - 1/m).$$

Hence $\Delta \le (1 - 1/m)\rho \le \rho$ and $\eta^2 \le L\rho$. For $h = f - g$, the mean of each h_i is zero. Equations (25) and (27) give the exact comparison and its error bound

$$S_\rho(f) - S_\rho(g) = -2D + \rho q + \mathcal{R}, \qquad 0 \le \mathcal{R} \le C_Q \rho D, \qquad C_Q = C_d M_+. \tag{31}$$

This bounds the perturbation's quadratic remainder; it does not estimate the difference of the two full higher-chaos tails separately.

Here the original mass error u in the repair lemma is zero. Thus (28)–(29) yield

$$\rho q \le (2 - \sigma)\rho J_{\mathrm{in}} + E_R D, \qquad E_R = 2C_J \varepsilon_R + 36(2K + 1)A_+^2 \varepsilon_R^2.$$

Choose a finite $R \ge 3$ with $E_R \le \sigma/4$, possible because $\varepsilon_R \to 0$. Choose any positive ρ_m^+ satisfying

$$\rho_m^+ \le \min\left\{\frac{1}{16}, \frac{\sigma}{16\beta_R}, \frac{\sigma}{4C_Q}, \frac{c_A^2}{576L}\right\}. \tag{32}$$

For every competitive assignment at $0 < \rho \le \rho_m^+$, $\eta \le c_A/24$, so the repair is available. Using (30) and $(1-\beta_R\rho)^{-1} \le 1 + 2\beta_R\rho$,

$$\rho q \le \left(\frac{2-\sigma}{1-\beta_R\rho} + E_R \right) D.$$

Since $4\beta_R\rho \le \sigma/4$, substituting this into (31) proves

$$S_\rho(f) - S_\rho(g) \le -\frac{\sigma}{4} D.$$

Strict ordering in (22) makes $D > 0$ unless $f = g$. Thus no distinct competitive assignment exists in dimension d. The finite-dimensional and equality transfer is completed in Section 4. The interval is closed at its upper endpoint because every defining bound in (32) is non-strict.

C The negative transfer with free common masses

We first work in dimension $d = m - 1$. A pair is competitive if $B_{-r}(f,h) \le B_{-r}(g,g)$, and its matching mass vector is denoted by p. Let $F = (f+h)/2$ and $U = \mathbf{1}/m$.

C.1 Uniform localization in the nearest frame

Parseval gives $\|b(f)\|_{\mathrm{F}}^2 \le \sum_i \mathrm{Var}(f_i) \le 1 - \|p\|^2 \le 1$, and likewise for h. The absolute value of the degree-at-least-two bilinear Hermite remainder is at most r^2, by Cauchy–Schwarz; the same bound holds for the reference. Competitiveness therefore implies

$$\|p - U\|^2 \le r(\langle b(f), b(h)\rangle_{\mathrm{F}} - w) + 2r^2 \le r + 2r^2, \qquad \langle b(f), b(h)\rangle_{\mathrm{F}} \ge w - 2r. \tag{33}$$

In particular all masses are positive if $r \le 1/(4m^2)$.

For every $\eta_* > 0$, there is a $\delta_* > 0$ such that all competitive pairs with $0 < r < \delta_*$ obey

$$\frac{1}{2} \sum_i \|F_i - g_i\|_1 < \eta_*, \tag{34}$$

where g is selected by *any* nearest regular centroid matrix to $b(F)$. To prove this uniformly, suppose otherwise and choose a violating sequence $r_n \downarrow 0$. Weak compactness of the bounded simplex-valued set in $L^2(\gamma_d)^m$ gives weak subsequential limits f_*, h_*. By (33), they are balanced and have centroid inner product at least w. The endpoint and Cauchy–Schwarz give the reverse chain

$$\langle b(f_*), b(h_*)\rangle \le \|b(f_*)\|\|b(h_*)\| \le w.$$

Equality forces identical centroid matrices of norm squared w, and endpoint equality gives $f_* = h_* = g_*$ for one ordered regular partition. Weak convergence to an indicator gives strong L^1 convergence, since $\sum_i \|f_{n,i} - g_{*,i}\|_1 = 2 - 2\sum_i \langle f_{n,i}, g_{*,i}\rangle \to 0$, and likewise for h_n, F_n. The centroids converge by weak continuity of first moments. Any nearest regular matrices c_n then satisfy $\|c_n - b(F_n)\| \le \|c_* - b(F_n)\| \to 0$. Their winning labels converge off the null hyperplanes of c_*, so their regular partitions converge in L^1. This contradicts the violation and proves (34).

C.2 The mixed comparison retains the mass penalty

Use a nearest regular frame for $b(F)$, and put

$$D = D_-(F), \quad z = \sqrt{D/r}, \quad \eta = \tfrac{1}{2}\sum_i \|F_i - g_i\|_1, \quad q = \|b(F) - c\|_{\mathrm{F}}^2, \quad u = \tfrac{1}{2}\|p - U\|_1.$$

The loss D_- is affine, so $2D = D_-(f) + D_-(h)$. Writing $q_f = f - g$ and $q_h = h - g$, the constant term in their mixed quadratic expansion is $\|p - U\|^2$. Its first-order term satisfies

$$\langle b(f) - c, b(h) - c\rangle = q - \tfrac{1}{4}\|b(f) - b(h)\|_{\mathrm{F}}^2 \le q.$$

The higher remainder has absolute value at most $C_Q r\sqrt{D_-(f)D_-(h)} \le C_Q r D$, where $C_Q = C_d M_-$. Consequently

$$B_{-r}(f,h) - B_{-r}(g,g) \ge \|p - U\|^2 + 2D - rq - C_Q r D. \tag{35}$$

No positivity of a negative-correlation higher-chaos term is assumed.

C.3 Repair and absorption of the unequal masses

Choose the localization cutoff (34) with $\eta_* = c_A/24$. The repair in Appendix A.5 is then available. Its mass imbalance satisfies both $v \le u + \varepsilon_R A_- z$ and $v \le \eta$. Using (28) and (29), $\eta \le A_- z$, and $(x+y)^2 \le 2x^2 + 2y^2$, we obtain

$$\begin{aligned} rq &\le (2-\sigma) r J_{\mathrm{in}} + B\sqrt{rD}\, u + 72(2K+1) r u^2 + \widetilde{E}_R D, \\ B &= 6A_-/c_A, \qquad \widetilde{E}_R = 2C_J \varepsilon_R + 72(2K+1)A_-^2 \varepsilon_R^2. \end{aligned} \tag{36}$$

Young's inequality gives $B\sqrt{rD}\, u \le \sigma D/8 + (2B^2/\sigma) r u^2$. Put $C_{\mathrm{mass}} = 72(2K+1) + 2B^2/\sigma$.

Choose finite $R \ge 3$ with $\widetilde{E}_R \le \sigma/8$. Then choose $r_m^- > 0$ small enough that

$$r_m^- \le \min\left\{ \frac{1}{16}, \frac{1}{4m^2}, \frac{\delta_*}{2}, \frac{\sigma}{32\beta_R}, \frac{\sigma}{8C_Q}, \frac{2}{mC_{\mathrm{mass}}} \right\}. \tag{37}$$

Here δ_* is the uniform cutoff just proved. Every entry is positive and depends only on m and its finite coercivity constant. For $0 < r \le r_m^-$, the compact comparison (30) and $(1-\beta_R r)^{-1} \le 1 + 2\beta_R r$ imply from (36)

$$rq \le (2 - 5\sigma/8) D + C_{\mathrm{mass}} r u^2.$$

Insert this into (35) and use $u^2 \le m\|p - U\|^2/4$ to get

$$B_{-r}(f,h) - B_{-r}(g,g) \ge \tfrac{1}{2}\|p - U\|^2 + \tfrac{\sigma}{2} D. \tag{38}$$

Thus a competitive pair has $p = U$ and $D = 0$. The strict gap (23) forces $F = g$; since g is an indicator and $F = (f+h)/2$, simplex-valuedness forces $f = h = g$ almost everywhere. This proves the full relaxed fixed-dimensional assertion. The remaining dimension reduction and recovery of fractional equality are given next in Section 4. No quantitative localization for unbalanced assignments was assumed: its uniform qualitative form was proved above.

D Proof lineage and review boundaries

The manuscript contains the covariance and endpoint proofs in full. Appendices A–C also contain the fixed-dimensional transfer proofs, including their constants and closed-endpoint choices. The analytic external inputs are the convex-support variance inequality [3, Proposition 2.1], the two dimension reductions with distinct mass scopes stated in Section 4, and, for the discrete corollary, the same-correlation transfer and plurality limit in [6]. No computational covariance cover is a premise of these results.

The assembly uses the following local sources, preserved separately in the review project's proof-source and audit directories. Hashes are SHA256; the companion reproduction manifest records the complete file inventory.

1. The all-dimensional covariance proof: `ALL_DIMENSIONAL_COVARIANCE_CONCAVITY.txt`.

   ```
   225db53f4489bf513926eead05cc5ec8
   c40962991246ac5036a897faa2d8130b
   ```

2. The general positive transfer: `FINITE_CELL_COERCIVITY_TO_NOISE_TRANSFER_20261030.txt`.

   ```
   1d951abe62965c31d3b8ce5209099d62
   167ed9fc9b554f0c91d4c057b9a3dff5
   ```

3. The general free-common-mass negative transfer: `FREE_COMMON_MASS_NEGATIVE_TRANSFER_AUDIT_20261031.txt`.

   ```
   9a6339628de87b4d0ae95526438caab8544
   da303bb2934c8494c204b33adc2ef
   ```

The older transfer notes have conditional historical status wording. Theorem 1.2 supplies precisely their hypothesis (1) for every fixed m; it is no longer an unproved endpoint assumption in the arguments presented here.

The accompanying records include the full cleanroom covariance audit, the separate end-to-end application audit, the exact scope note, and the root implication note. These are internal AI-agent audits; no external reviewer is implied. The independent geometry checker verifies the facet-basis construction, the full chain factorization, and the redundant gauge equations on rational fixtures in dimensions two through seven. Its 96 exact checks support the algebra but do not replace the uniform proof or the analytic hypotheses. Earlier frozen four-cell releases retain their original statements and numerical intervals; they are not dependencies of the new all-cell analytic covariance proof.

This project is a local review artifact. Internal AI-agent mathematical audits and reproducibility checks are distinguished from external peer review. No author or affiliation has been assigned, and no publication, external contact, or priority claim is made. The main conclusions are for each fixed cell count near independence; full positive and negative correlation ranges remain open.

References

[1] S. Heilman, *Euclidean Partitions Optimizing Noise Stability*, 2014. arXiv:1211.7138v2. Section 8, Conjecture 3 and the preceding discussion.

[2] S. Heilman, *On the Propeller Conjecture*, 23 October 2026. arXiv:2610.32158v1. Introduction following Theorem 1.3.

[3] S. G. Bobkov and M. Ledoux, *From Brunn–Minkowski to Brascamp–Lieb and to logarithmic Sobolev inequalities*, Geometric and Functional Analysis **10** (2000), 1028–1052. Proposition 2.1, printed p. 1031. Author-hosted full text.

[4] S. Heilman and A. Tarter, *Three Candidate Plurality is Stablest for Small Correlations*, Forum of Mathematics, Sigma **9** (2021), e65. arXiv:2011.05583v2. Theorem 1.9.

[5] S. Heilman, *Hyperstable Sets with Voting and Algorithmic Hardness Applications*, 2022. arXiv:2209.11216v1. Problem 7.1 and Theorem 7.9.

[6] M. Isaksson and E. Mossel, *Maximally Stable Gaussian Partitions with Discrete Applications*, 2009. arXiv:0903.3362v3. Theorem 7.1 and Lemma 7.3.